AI-Driven NPCs Enhancing Player Challenges and Skill Development in Games

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Figure 1: "Down the Rabbit Hole" game and AI NPC conversational agent.

Abstract

Non-player characters (NPCs) play a crucial role in shaping player experiences in narrative- driven games. This study investigates the potential of AI-driven NPCs to elevate player challenges, foster skill development —such as critical thinking, emotional regulation, and strategic decision-making— and deepen emotional engagement. Focusing on adaptive conversational agents, we explore how dynamic, context-aware dialogue systems can enhance player immersion and agency. In our experimental 3D RPG prototype, Down the Rabbit Hole, AI-powered NPCs respond organically to player input, influencing gameplay through evolving relationships and personalized guidance. Through a small-scale exploratory user study, we observe how AI-driven interactions impact player progression, perceived challenge, decision-making, and emotional connection, offering preliminary insights into the design of more responsive and meaningful narrative experiences.

CCS Concepts

• **Applied computing** → **Computer games**; • **Human-centered computing** → **HCI design and evaluation methods**.

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Keywords

AI-driven NPCs, Role-playing games (RPGs), conversational agents, game experience, player engagement, Human-AI interaction in games

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1 Introduction

Role-playing games (RPGs) have consistently attracted players with their expansive worlds, immersive storytelling, and the promise of meaningful player-driven choices [5]. Whether they take place in fantasy worlds or speculative futures, RPGs are designed to give players the freedom to create their own stories. However, RPGs still face challenges such as rigid morality systems, repetitive quests, and overly simple difficulty levels [20], which reduce narrative depth and player engagement [5, 21]. Additionally, research on player choices in RPGs show that having control over the story greatly affects how engaged and satisfied players feel. Players usually make decisions that align with their roles, which is known as the "Mimesis Effect" [6]. The choices that matter, especially those that involve moral, social and outcome-related issues, are especially important [13], while branching narratives and increased decision points have been shown to enhance the player experience [16, 17]. It has been found that choice structure strongly influences player agency, immersion, and emotional investment [15, 23] while

understanding player decision-making processes and the multi-level impact of choices on gameplay is essential for designing more impactful narrative-driven games [26].

By using conversational AI, game developers can create Non Player Characters (NPCs) that have changing personalities, deep emotions, and behaviors suitable for different situations[30]. This can enhance players' feelings of connection and thoughtfulness during their interactions in the game. In this article, we explore how AI can change the way players engage with NPCs and have a crucial role in story-driven games. While RPGs typically rely on preset dialogues and fixed options, recent advances in machine learning, player modeling, and natural language processing can make NPCs more dynamic, emotionally aware, and better at understanding their environment. As a result, these new ideas raise a larger question. Can AI not only make NPCs more realistic but also change how a branching story develops based on how players interact with it? To this end, we introduce "Down the Rabbit Hole," a prototype 3D first-person game in which players engage with "Alice," an AI-driven conversational agent that changes its behavior and advice depending on the player's answers, tone of voice, emotional signals, and growing connection during the interaction.

As a result, we focused our study on three main research questions (RQ):

- RQ1: How does the presence of an AI-driven NPC affect players' perception of challenge and decision-making in a narrative RPG?
- RQ2: Does the adaptive conversational behavior of the NPC affect player engagement and emotional connection?
- RQ3: Can such dynamic dialogue experiences foster learning or strategic thinking in games?

2 Related Work: AI-driven NPCs

The NPCs led by the AI are increasingly characterized by their ability to learn from players' strategies and adapt accordingly. This adaptability is often obtained through automatic learning algorithms that allow them to change their tactics in real time, providing a shaded game experience [14]. Unlike traditional NPCs, which follow predictable patterns, these characters enhanced by the AI can analyze the player's behavior and respond with greater refinement, thus increasing the level of difficulty of the game. Research shows that the use of adaptive artificial intelligence techniques can significantly raise the challenge presented to the players, contributing to an advanced overall gaming experience [10]. A fundamental area of exploration is the concept of dynamic adaptation of the difficulty (DDA). DDA algorithms allow games to alter difficulty based on players' performance, perfecting the challenge to maintain involvement without causing frustration [3]. When NPCs can adapt their difficulty through learning mechanisms, such as reinforcement learning or genetic algorithms, players are required to develop new strategies to overcome increasingly sophisticated challenges [1]. This evolution interaction encourages not only the adaptability from the players, but also promotes the development of skills while players meet different tactical commitments [27].

The strategic challenges presented by NPCs are manifested in various game genres, in particular RPGs where they simulate man-similar behaviors, creating more realistic scenarios in which players must think critically of their next moves. For example, biologically used algorithms used in the NPC movement improve unpredictability and realism [24], forcing players to continuously adapt their strategies. This involvement can encourage a sense of realization and mastery while players learn to navigate in complex game environments with intelligent opponents [8]. Moreover, the impact of AI-driven NPCs naturally extends to the player's experience, transforming games from a passive activity into a highly interactive one. Studies have shown that players report higher satisfaction levels when they face intelligent NPCs that listen dynamically [22]. These NPCs challenge players on several levels, promoting cognitive involvement and the ability to resolve the problems that are crucial for progress within the game [18]. Despite advantages, implementing AI-driven NPCs in the game design process brings new challenges. The developers must balance the complexity of AI behavior with the game function, ensuring that the NPCs improve the gameplay without overwhelming players [29]. In addition, there is an ongoing discussion on the ethical implications of AI advanced in the games, in particular linked to the autonomy and equity of the players [28]. While the gaming industry continues to evolve, the future of the NPCs led by the AI promises even deeper impacts on the involvement and experience of the players. So, a better understanding of AI applications in games will probably lead to even more sophisticated NPC projects that improve the difficulty and satisfaction of the players [7]. Exploring these dimensions, future research can encourage the development of solid framework that facilitate significant interactions between players and intelligent game entities [11, 25]. Additionally, models based on affective computing and player psychology provide prove extremely valuable for designing emotionally engaging NPC interactions. For instance, Csikszentmihalyi's Flow theory [4] emphasizes the importance of utilizing challenge and skill to maintain player immersion and focus. Similarly, Narrative Transportation Theory [9] suggests that emotional investment and mental imagery while exploring a story, can deepen narrative immersion. Furthermore, affective computing frameworks [19] highlight how systems that interpret and adapt to emotional cues—such as tone or sentiment—(using machine-analyzable data from the player, usually in real time) can enhance believability and presence in human-computer interaction. These models offer theoretical foundation for the design of emotionally responsive AI-driven NPCs in our system.

3 The Game: Down the Rabbit Hole

3.1 Game Concept and Design Goals

"Down the Rabbit Hole" is a first-person narrative RPG set in a twisted, darker version of Wonderland. Players take on the role of a determined protagonist on a dangerous mission to rescue their kidnapped sister, Alice, from the malevolent Queen of Hearts. Players must navigate branching dialogue with twisted reimaginings of their favorite characters, discover cryptic lore, and survive the Queen's ruthless dominion through choice-driven encounters and psychological, often perilous, exploration. Alice is the game's emotional and narrative anchor. She is not only the player's younger sister but also the first implemented AI-powered NPC. Alice exists as a dynamic character capable of fluid, context-sensitive conversation. Though she doesn't yet respond to an affinity system, her

role is deeply integrated into the plot, acting as both guide and mystery. The core gameplay loop revolves around exploration, interaction, and dialogue-driven progression. Instead of traditional quest markers and exposition dumps, the player must engage in meaningful conversations to uncover Wonderland's fractured politics, emotional tensions, and hidden dangers. These conversations are not about linear progression—they are the story. For example, during one interaction the player asks, "What is the huge house over there?" and Alice replies, "That's our family's estate [Player Name]. It has been in our family for generations. It's grand but I prefer the peacefulness of the garden." This demonstrates how Alice not only provides the player with lore, but subtly reveals her personality and preferences, setting the emotional tone of the conversation.

The design philosophy is anchored around three main pillars: immersion, meaningful choice, and emotional resonance. Our key goals for "Down the Rabbit Hole" are:

- Create Emotionally Responsive AI Characters
- Develop a Lightweight Affinity System
- Treat Dialogue as Gameplay
- Enable Replayable Narrative Structures
- Maintain Narrative Cohesion in Generative Systems

These goals are showcased in how players experience Alice's personality and emotional depth. For example, when asked "What are you reading, Alice?", she responds warmly with, "Gulliver's Travels... It's a fascinating adventure story." and even offers to explain it in detail.

3.2 Implementation and Adaptive Narrative Structure

"Down the Rabbit Hole" implemented using UE 5.3, and the VaRest plugin for real-time HTTP communication with OpenAI's API. Each NPC is contextually defined through a premade system prompt that includes: character backstory, goals, personality traits, cur- rent emotional state, and narrative relevance. This framing ensures consistent, character- true responses, even in unpredictable conversations. Alice, the first and currently the only implemented character, does not utilize the affinity system. The branching system is modular, with a scalable foundation that begins with simple affinity scoring per interaction. While Alice remains a fixed character, other NPCs can be designed to dynamically react to the player's behavior through a dialogue-specific affinity model. The NPC Conversation Structure as shown in Figure 2 operates as follows:

- Initiation: NPCs or players may begin conversations, depending on the narrative context.
- Information Flow: NPCs provide guidance, world-building details, and emotionally shaded commentary based on their character perspective.
- Minimum Dialogue Exchanges: Each interaction includes a set number of exchanges to ensure depth and meaningful engagement.
- Embedded Personality Queries: NPCs could ask subtle questions, or simply judge by the player's lines throughout the exchange, evaluating the player's personality and values based on the responses.
- Real-Time Affinity Scoring: At the conclusion of a conversation, the NPC calculates an affinity score between 0 and 1 based on shared values, player tone, and the perceived alignment.
  - Above 0.5: The NPC offers accurate, helpful guidance—resulting in safer, more optimal routes or choices.
  - Below 0.5: The NPC deliberately misguides the player, increasing environmental hazards or narrative complications—without revealing the deception.

It is important to note this affinity scoring is currently session-based only and does not add up over time. There is no memory between conversations, each interaction is judged on its own. The affinity percentage affects only the immediate outcome of the current dialogue. Future iterations could extend the affinity system to track choices across time, adding long- term consequences, character memory, and cross-NPC influence, by creating "tags" for the player. Eventually, characters will reference past actions, react differently on re-peat encounters, and unlock or restrict paths based on trust built (or lost) through the game. For instance, in one exchange the player asks, "Is this dress new?" and Alice responds, "Mother bought my dress from the seamstress in the village. She said the color reminded her of the turquoise waters at the seaside. I feel like a princess when I wear it." Moments like these showcase how Alice's personality, complex emotional expression, and her knowledge of situation context, support a naturalistic affinity modeling. Player choices also influence emotional dynamics. In one case, when the player expresses interest in Alice's reading, she eagerly explains the plot of Gulliver's Travels, ending with, "Wouldn't that be funny, [Player Name]?"—inviting the player to conversate. But when asked repeatedly about her identity, Alice playfully but firmly insists, "I'm your younger sister, Alice… don't be silly." These interactions reflect an emerging personality and affinity profile, where Alice responds warmly to curiosity, but becomes more defensive or corrective when questioned too bluntly. In future iterations, we aim to extend this system to track long-term player-NPC relationships. A more advanced affinity system will allow NPCs to remember prior interactions, reference past events, and exhibit evolving trust levels—deepening player investment and enabling persistent consequences across the narrative.

4 Player Experience and Evaluation

The player evaluation aimed to explore how the adaptive behavior of AI-driven NPCs, particularly Alice, influences player perceptions of emotional presence, intelligence, challenge, and overall enjoyment or frustration. So, we conducted a small-scale user study using, *Down the Rabbit Hole*. The study involved 7 undergraduate students (3 male, 4 female) from the "Game Design and Development" course at our university. Each participant played a 15–20 minute segment that focused on a free-form conversation with Alice. After the session, the participants completed two validated instruments:

- **Godspeed Questionnaire** [2]: assessing Animacy, Intelligence, and Likeability
- **Game Experience Questionnaire (GEQ)** [12]: assessing Immersion, Flow, Challenge, and Tension

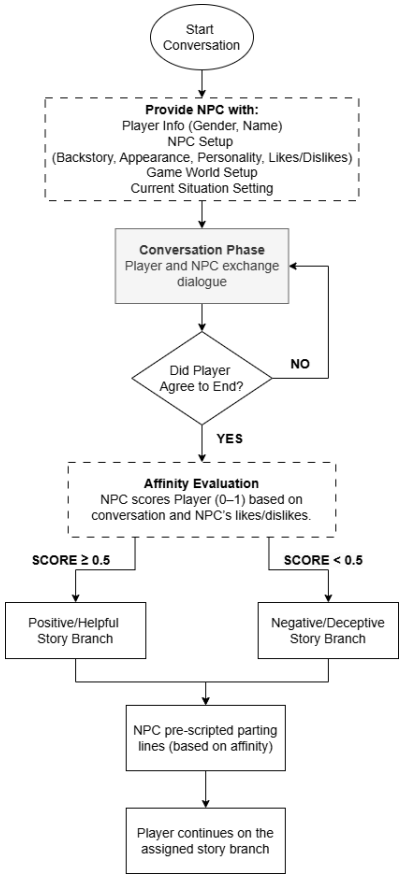

Figure 2: NPC interaction and affinity scoring flow in *Down the Rabbit Hole*. The diagram illustrates how conversations are initiated, how player choices are evaluated, and how NPC behavior is influenced by affinity scoring conducted at the end of each dialogue.

Optional interviews and logs of Alice-player conversations allowed for further qualitative and behavioral analysis.

In Table 2, we present a summary of our measurement tools and hypotheses. We calculated composite scores in Table 1 by averaging the Animacy and Intelligence subscales from the Godspeed Questionnaire to represent perceived emotional intelligence. For the Game Experience Questionnaire (GEQ), Immersion and Tension subscale scores were averaged separately. Challenge scores were used directly from the GEQ. Additionally, a Pearson correlation analysis revealed a significant positive relationship between Godspeed Likeability and GEQ Enjoyment ($r = 0.67$, $p < 0.05$), indicating that participants who rated Alice as more likable also tended to report greater enjoyment.

Participants had short post-play interviews that were thematically analyzed using affinity mapping. Key questions focused on perceived realism, emotional connection, and dialogue experience. Three major thematic groups emerged:

- Emotional Presence and Realism: Several participants noted Alice's responses felt "genuine" or "strangely human" and "in-character," pleased with the NPC's adaptive language.
- Confusion from Open Format: Some reported difficulty handling the conversation: "I didn't know what I could say to move on, or what information I should learn."
- Trust and Personality Cues: Players mentioned trying to guess Alice's mood, intent and role in the story: "I felt like she was hiding something, so I tried to be careful."

Overall, participants responded positively to the adaptive AI-driven NPC. Alice was perceived as intelligent and emotionally engaging, especially when she referenced earlier parts of the conversation, creating a sense of memory. Players noted the freedom of expression as empowering, though some struggled with the lack of conversational scaffolding. This highlights the trade-off between open-ended interaction and player guidance. Despite its limitations (e.g., no long-term memory or quest outcomes), the system showed strong potential to enhance narrative immersion and strategic dialogue thinking. Participants rated conversational agents like Alice as emotionally engaging, which may increase narrative engagement, challenge perception, and emotional presence in RPG contexts. However we must take into account that this study did not include a control group, as it was designed as an exploratory pilot to assess feasibility and initial player impressions.

5 Discussion and Conclusion

AI-driven NPCs have the potential to transform RPG experiences by enabling dynamic, personalized conversations. Unlike static dialogue trees, these agents can respond contextually to player input, creating interactions that feel more lifelike and emotionally engaging (supporting H1 and H2, tied to RQ2). This adaptivity may increase replayability, as some participants described an interest playing again to explore different conversational strategies or narrative paths. While replay behavior was not directly measured, players expressed curiosity about how alternate inputs might produce different responses, suggesting perceived narrative variability (RQ3). Our findings suggest that players experienced the open conversational format as both enriching and challenging (supporting H4, RQ1). While story-driven players appreciated the ambiguity and emotional presence of Alice, some goal-oriented users found the lack of dialogue scaffolding frustrating— highlighting the balance required between freedom and structure.

Nonetheless, challenges remain and of course our study has limitations, including a small sample size (N=7), a short session duration (15–20 minutes), and the absence of a control condition. As such, findings should be interpreted as preliminary and should not be generalized. Additionally, technical limitations of the AI system, including lack of long-term memory and limited affinity modeling, may impact NPC behavior. AI should enhance, not replace, the narrative design that defines RPGs. Players responded positively to dynamic NPC behavior, which increased immersion and engagement. Future work should explore broader adaptivity, such as emotional memory and behavior-based relationship evolution. Moving forward, NPCs could form opinions shaped not only by spoken interactions but by observing a player's moral decisions, social behavior, and movement patterns throughout the game world. This would create a more deeply reactive and emotionally complex environment, where relationships with NPCs grow, change, and sometimes deteriorate based on the player's actions.

Table 1: Measurement tools and variable interpretation for exploratory study. Subscale scores were averaged to yield final metrics. Higher values reflect stronger agreement.

Hypothesis	Measurement Tool	Scale or Metric
H1: Alice perceived as emotionally intelligent	Godspeed Questionnaire	Composite score from Animacy and Intelligence subscales (1–5 scale)
H2: Increased immersion and tension	GEQ	Immersion & Tension subscales (0–4 scale, averaged separately)
H3: Emotional presence correlates with enjoyment	Correlation Analysis	Likeability (Godspeed) × Enjoyment (GEQ)
H4: Open format increases perceived challenge	GEQ	Challenge subscale (0–4 scale)

Table 2: Summary of observed ratings and corresponding measurement items. Reported means are descriptive only, not confirmatory.

Hypothesis	Observations	Evidence	Relevant Scale Items
H1: Higher emotional intelligence	Participants rated high	Composite score from Animacy and Intelligence: M = 4.3/5 (SD = 0.6)	Godspeed: Animacy, Intelligence
H2: Immersion and uncertainty	Mixed responses	Immersion M = 4.1, Tension M = 3.5	GEQ: Immersion, Tension
H3: Presence & Enjoyment correlation	Correlation observed	$r = 0.67,\ p < 0.05$	Godspeed Likeability × GEQ Enjoyment
H4: Conversational challenge	Participants rated game as challenging	Challenge M = 3.9/5	GEQ: Challenge

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